

Human Expert Evaluation: Synthesis of Findings

Overview

Blinded evaluations were conducted with three human domain experts across three domains. Reviewers received paired outputs (Condition A and Condition C) with randomized labels and no indication of which was generated by which method or that the study concerned a pipeline intervention.

AUSA 2025 (Defense Policy)

Reviewer 1: Defense Policy Expert (n=2 rounds)

Round 1 (outside-view pipeline calibration):

Reviewer scored both outputs on 5 practitioner-oriented dimensions (1–5 scale). Extensive qualitative feedback revealed that the pipeline's interpretive additions were perceived as obvious to a domain insider:

"Of course AUSA does advocacy. Of course nobody says China is the adversary. Of course they don't talk about DOGE and piss off the admin. Of course they don't interview soldier spouses."

The reviewer's overall assessment: if the interpretation is bad, "she only wants the facts, and the naive one sticks to the facts." The pipeline's assumption surfacing—its strongest dimension by AI judge scores—was perceived as stating field-obvious truths rather than generating novel insight. The gaps flagged by the pipeline "aren't the ones that an expert would want flagged from the material."

Round 2 (inside-view pipeline calibration):

After recalibrating the pipeline's operations to assume domain expertise in the reader, the reviewer found the outputs "much better overall" but noted persistent issues: "yes the AUSA news corpus is propaganda, yes everyone does what the generals say because the org is hierarchical, no nobody is going to name the adversary." The reviewer preferred the fine-grained pipeline variant, which aligned with AI judge preferences. However, the core critique persisted: corpus-structural assumptions obvious to any reader of the source material were still being surfaced as insights.

AAHS 2026 (Hand Surgery)

Reviewer 2: Hand Surgeon

Reviewer read both outputs and reported: "honestly the 2 were similar. It will take a deeper dive to discern major differences."

On the content itself: "Major discussion topics: Declining reimbursement, peri-operative medications, WALANT economics, IM fixation and Socioeconomic impacts. Nothing new or informative for me. I tend to be ahead of the curve when it comes to emerging technologies. It serves as a reinforcement of what I presumed to be, so in that sense it is informative."

The reviewer confirmed that the corpus captured the right themes but found neither output provided information beyond what he already knew. The synthesis was accurate but not additive for an expert reader.

ICLR 2025 (Machine Learning)

Reviewer 3: Active ML Researcher

Reviewer read both outputs and reported: "One isn't obviously 'better' than the other." Differences were characterized as "mostly about taste" once factual accuracy was assumed: "beyond me fact checking things and assuming the model counted things correctly, etc the differences are mostly about taste."

Detailed feedback on the pipeline output (Condition C):

Valued:

- "I liked the way that the prose for shape of the field was structured, that it provided the big themes before diving in."
- "None of its contents were thematically surprising to me, which built trust."
- Cross-cutting themes section and concise bullet-list formatting for trajectory predictions.

Critiqued as obvious or unhelpful:

- "The trajectory and implications section... these were mostly obvious conclusions."
- "The methodological observations are pretty obvious. It'd be more interesting to contrast with the past and flag opportunities."
- "If I were an 'active' researcher I don't know if I'd have found them helpful beyond it being a cute summarization."

- On the "What is Missing" section: "I can see why it cared about environmental and energy costs, etc. But not sure if ICLRs ever spent much energy on that stuff other than performativity."

Wanted but absent:

- Hyperlinks to specific papers or OpenReview IDs.
- Highlighting of outlier papers: "I'd have wanted it to highlight the 'weird' and therefore possibly more interesting papers."
- Personalized relevance filtering: "The even more powerful thing would be to let each researcher rapidly shape the relevance landscape of the thematic analysis based on their research interests."
- Quality-based filtering by institution, methods rigor, etc.
- Temporal comparison: "Rather than underrepresented areas, it'd have been more interesting to contrast the thematic analysis of this year with last year."
- Normative transparency: "Notable individual contributors should have offered a rationale for why those specific ones were picked. Or more specifically, what overarching normative principles it used to pick the notable ones."
- Raw intermediate artifacts over polished synthesis: "It'd almost be helpful to have excavate or ramify outputs on the corpus as a whole. I.e. assumptions or predictions."

Detailed feedback on the baseline output (Condition A):

- "The executive summary in 895 isn't helpful. We should just get rid of it."
- "I like that the other pdf [C] provided bullets of the thematic patterns before diving into them."
- Similar critique about clustering transparency and missing normative justification.

Overall assessment: "The most value from this PDF was the thematic analysis assuming it was correct." And critically: "This is a sort of 'summarization' task which is inherently underspecified. So different people will have wildly different preferences than what I'm saying here."

Cross-Reviewer Synthesis

Three patterns emerge from human expert evaluation that diverge sharply from AI judge assessments:

1. **Domain experts cannot reliably distinguish pipeline from baseline.** No reviewer

identified one output as clearly superior overall. Differences were characterized as matters of “taste” (Reviewer 3), requiring “a deeper dive to discern” (Reviewer 2), or favoring the baseline’s factual density over the pipeline’s interpretive additions (Reviewer 1).

2. The pipeline’s strongest AI-judge dimension is its weakest human dimension.

Assumption surfacing—the largest and most consistent gain by AI judge scores (+1.33 to +2.00 across domains)—was consistently perceived by domain experts as stating the obvious. Field-structural assumptions (“AUSA does advocacy”), corpus-structural assumptions (“the corpus is propaganda”), methodological observations (“many papers lack rigorous evaluation”), and thematic trajectories (“mostly obvious conclusions”) were all judged as known truths rather than novel insights. As Reviewer 2 summarized: the synthesis “serves as a reinforcement of what I presumed to be.”

3. Experts want navigation, not synthesis. Rather than a better essay about the field, reviewers wanted: personalized relevance filtering, temporal comparisons with prior years, anomaly highlighting (“weird” papers), quality-based filtering, networking recommendations, and access to raw analytical artifacts rather than polished reports. Reviewer 3 explicitly requested the pipeline’s intermediate outputs (excavate, ramify) as standalone products rather than inputs to synthesis. The synthesis product solves a problem experts do not report having; the analytical intermediates may be closer to what they actually need.

These findings suggest that AI judges and human domain experts evaluate synthesis quality along fundamentally different axes. AI judges reward epistemic structure—stratified evidence, surfaced assumptions, falsifiable predictions. Human experts reward decision-relevant novelty—information they didn’t already know, presented in forms they can act on. The pipeline reliably produces the former but not the latter. This divergence raises questions about the validity of LLM-as-judge evaluation for synthesis tasks and suggests that epistemic depth, as currently operationalized, may be orthogonal to expert utility.